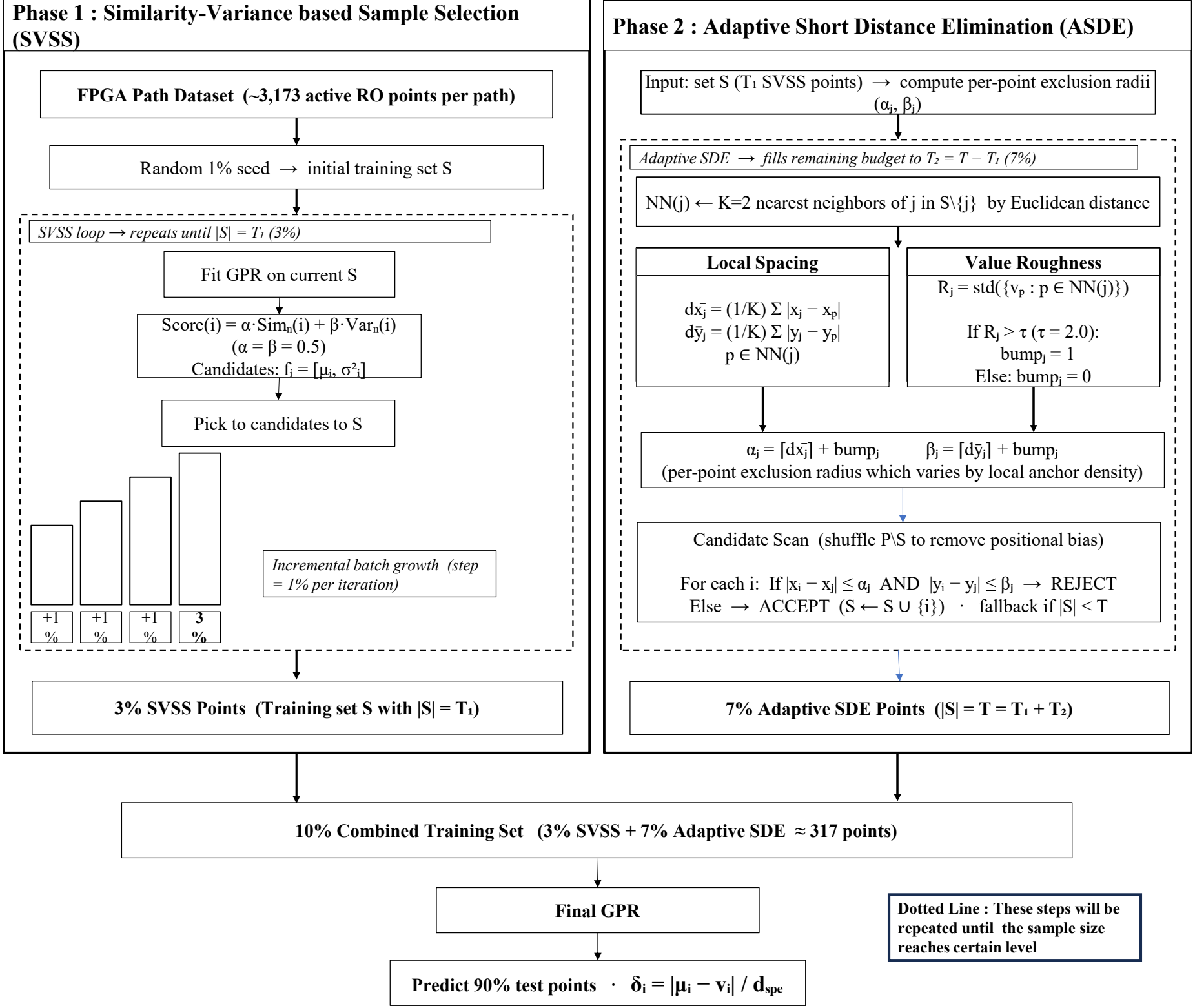

Phase 1 : Similarity-Variance based Sample Selection (SVSS)
FPGA Path Dataset (~3,173 active RO points per path)
Random 1% seed → initial training set S
SVSS loop → repeats until |S| = T1 (3%)
Fit GPR on current S
Score(i) = α·Simn(i) + β·Varn(i)
(α = β = 0.5)
Candidates: fi = [μi, σ2i]
Pick to candidates to S
+1 %
+1 %
+1 %
3 %
Incremental batch growth (step = 1% per iteration)
3% SVSS Points (Training set S with |S| = T1)
Phase 2 : Adaptive Short Distance Elimination (ASDE)
Input: set S (T1 SVSS points) → compute per-point exclusion radii (αj, βj)
Adaptive SDE → fills remaining budget to T2 = T − T1 (7%)
NN(j) ← K=2 nearest neighbors of j in S\{j} by Euclidean distance
Local Spacing
dx̄j = (1/K) Σ |xj − xp|
dȳj = (1/K) Σ |yj − yp|
p ∈ NN(j)
Value Roughness
Rj = std({vp : p ∈ NN(j)})
If Rj > τ (τ = 2.0):
bumpj = 1
Else: bumpj = 0
αj = ⌈dx̄j⌉ + bumpj βj = ⌈dȳj⌉ + bumpj
(per-point exclusion radius which varies by local anchor density)
Candidate Scan (shuffle P\S to remove positional bias)
For each i: If |xi − xj| ≤ αj AND |yi − yj| ≤ βj → REJECT
Else → ACCEPT (S ← S ∪ {i}) · fallback if |S| < T
7% Adaptive SDE Points (|S| = T = T1 + T2)
10% Combined Training Set (3% SVSS + 7% Adaptive SDE ≈ 317 points)
Final GPR
Predict 90% test points · δi = |μi − vi| / dspe
Dotted Line : These steps will be repeated until the sample size reaches certain level